

ASSIGNMENT-6

- ① 6 data points
↳ 5 intervals.

$$f(x) = a_i x^2 + b_i x + c_i.$$

$$\text{At } T=0; \rightarrow a_1 0^2 + b_1 0 + c_1 = 14.621 \quad \text{--- (1)}$$

$$\boxed{c_1 = 14.621}$$

Spline No

- ② @ $T=40 \rightarrow 1600a_5 + 40b_5 + c_5 = 6.418.$
- ③ @ $T=4 \rightarrow 64a_1 + 8b_1 + c_1 = 11.843 \quad \text{①}$
- ④ @ $T=8 \rightarrow 64a_2 + 16b_2 + c_2 = 11.843. \quad \text{②}$
- ⑤ @ $T=16 \rightarrow 256a_2 + 16b_2 + c_2 = 9.870 \quad \text{②}$
- ⑥ @ $T=16 \rightarrow 256a_3 + 16b_3 + c_3 = 9.870 \quad \text{③}$
- ⑦ @ $T=24 \rightarrow 576a_3 + 24b_3 + c_3 = 8.418. \quad \text{③}$
- ⑧ @ $T=24 \rightarrow 576a_4 + 24b_4 + c_4 = 8.418. \quad \text{④}$
- ⑨ @ $T=32 \rightarrow 1024a_4 + 32b_4 + c_4 = 7.305. \quad \text{④}$
- ⑩ @ $T=32 \rightarrow 1024a_5 + 32b_5 + c_5 = 7.305. \quad \text{⑤}$

$$f'(x) = 2ax + b.$$

$$\textcircled{a} T=8; \rightarrow 16a_1 + b_1 = 16a_2 + b_2$$

$$16(a_1 - a_2) = b_2 - b_1.$$

$$16(a_1 - a_2) + b_1 - b_2 = 0$$

$$\textcircled{c} T=16 \Rightarrow 32a_2 + b_2 = 32a_3 + b_3$$

$$32a_2 + b_2 - 32a_3 - b_3 = 0$$

$$\textcircled{d} T=24 \Rightarrow 48a_3 + b_3 = 48a_4 + b_4$$

$$48a_3 + b_3 - 48a_4 - b_4 = 0$$

$$f''(x) = b$$

$\hookrightarrow f''$ at first point we are assuming $= 0$

$$\boxed{a_1 = 0} \quad \textcircled{-15}.$$

$$\rightarrow \textcircled{a} T=32 \rightarrow 64a_4 + b_4 = 64a_5 + b_5$$

$$\Rightarrow 64a_4 + b_4 - 64a_5 - b_5 = 0.$$

$$\textcircled{a} \quad f[x_1, x_0] = \frac{11.843 - 14.621}{8 - 0} = -0.34725$$

$$f[x_2, x_1] = \frac{9.870 - 11.843}{16 - 8} = -0.246625$$

$$f[x_3, x_2] = \frac{6.4148 - 9.870}{24 - 16} = -0.1815$$

↓
as soon we
can continue.

$$\boxed{\begin{aligned} f[x_i, x_j] \\ = \frac{f(x_i) - f(x_j)}{x_i - x_j} \end{aligned}}$$

second dividend
diff.

$$f[x_2, x_1, x_0] = \frac{-0.246625 - (-0.34725)}{16 - 0} = 0.00625906$$

$$f[x_3, x_2, x_1] = \frac{-0.1815 - (-0.246625)}{24 - 8} = 0.00407031$$

Third dividend diff:

$$f[x_3, x_2, x_1, x_0] = \frac{0.00407031 - 0.00625906}{24 - 0} = -0.00091245$$

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fourth diff

$$f[x_4, x_3, x_2, x_1, x_0] = \frac{-0.00005294 - (-0.00009245)}{32-0} \\ = 0.00000104.$$

fifth diff

$$f[x_5, x_4, \dots, x_0] = \frac{0.00000065 - 0.00000104}{40-0} \\ = -0.00000001.$$

$$O(t) = 14.621 - 0.34725(t) + 0.00628906(t)(t-8) - \\ 0.00009245(t)(t-8)(t-16) \\ + 0.00000104(t)(t-8)(t-16)(t-24) \\ - 0.00000001(t)(t-8)(t-16)(t-24) \\ (t-32).$$

② so, we have even number of points. ($T=114$).

$$y_t = a_0 + \sum_{k=1}^{k_{\max}} \left[a_k \cos\left(\frac{2\pi k t}{T}\right) + b_k \sin\left(\frac{2\pi k t}{T}\right) \right]$$

$$k_{\max} = \frac{T}{2} = \frac{114}{2} = 57.$$

$$a_0 = \frac{1}{T} \sum_{t=1}^T y_t.$$

$$a_k = \frac{2}{T} \sum_{t=1}^T y_t \cos\left(2\pi \frac{k t}{T}\right)$$

$$b_k = \frac{2}{T} \sum_{t=1}^T y_t \sin\left(2\pi \frac{k t}{T}\right)$$

① $k=10$

② $a_{10} = \frac{2}{114} \left[y_1 \cos\left(\frac{2\pi(10)(1)}{114}\right) + y_2 \cos\left(\frac{2\pi(10)(2)}{114}\right) + \dots + y_{114} \cos\left(\frac{2\pi(10)(114)}{114}\right) \right]$

$$a_{10} = \frac{2}{114} \left(2704899 \cos\left(\frac{20\pi}{114}\right) + 3105421 \cos\left(\frac{40\pi}{114}\right) + \dots \right)$$

$$b_{10} = \frac{2}{114} \left[2704899 \sin\left(\frac{20\pi}{114}\right) + 3105721 \left(\frac{40\pi}{114}\right) \right]$$

$$\text{Power} = a_0^2 + b_0^2 = (221181.94)^2 + (-35187.76)^2 \\ = 5.02 \times 10^{10}$$

③ $k = 39$

$$a_{39} = \frac{2}{114} \left(y_{\cos}\left(\frac{2\pi(39)(1)}{114}\right) + y_2 \cos\left(\frac{2\pi(39)(2)}{114}\right) \right)$$

$$b_{39} = \frac{2}{114} \left(y_1 \sin\left(\frac{2\pi}{114}\right) + y_2 \sin\left(\frac{156\pi}{114}\right) \right)$$

$$a_{39}^2 + b_{39}^2 = (-19313.05)^2 + (158754.82)^2 \\ = 3.95 \times 10^{10}$$

② $k = 2,$

We can write same as above

$$\text{Power} = a_2^2 + b_2^2 = (38098.26)^2 + (190398.73)^2 \\ = 3.77 \times 10^{10}$$

$a_0 = 2121826.79$

Rank 1: $k=10$, Frequency= 0.0877 , Power= $5.02e+10$

Rank 2: $k=39$, Frequency= 0.3421 , Power= $3.95e+10$

Rank 3: $k=2$, Frequency= 0.0175 , Power= $3.77e+10$

$\rightarrow$ for (11.4 yrs)
for $k=10$ $\Rightarrow \frac{10}{114} = 0.0877$ cycles/year (first dominant)

second dominant: $k=39 \Rightarrow f = \frac{39}{114} = 0.3421$ cycles/year

(for 2.92 yrs) $\leftarrow$
 $k=2 \Rightarrow f = \frac{2}{114} = 0.0175$ cycles/year
(57 years)